

CSCC63 – WEEK 5 TUTORIALS

There are several questions here, you aren't expected to finish all of them today. But they'll be good practice for you.

1. Find a set Y_1, Y_2, Y_3, Y_4 , and Y_5 of languages such that:

- $Y_1 \subseteq Y_2 \subseteq Y_3 \subseteq Y_4 \subseteq Y_5$,
- Y_3 is decidable,
- Y_2 is recognizable but not co-recognizable,
- Y_4 is co-recognizable but not recognizable, and
- Y_1 and Y_5 are neither.

2. Suppose that $A \leq_m B$ and $B \leq_m C$. Prove that $A \leq_m C$.

3. Recall the four languages from last week:

- (a) $X_1 = \{\langle M \rangle \mid \text{There is some } w \in \{0, 1\}^* \text{ such that } M \text{ accepts both } w \text{ and } ww\}$.
- (b) $X_2 = \{\langle M \rangle \mid \text{There is some } w \in \{0, 1\}^* \text{ such that } M \text{ accepts } w \text{ but loops on } ww\}$.
- (c) $X_3 = \{\langle M \rangle \mid \text{Whenever } M \text{ accepts } w \in \{0, 1\}^*, M \text{ loops on } ww\}$.
- (d) $X_4 = \{\langle M \rangle \mid \text{Whenever } M \text{ loops on } w \in \{0, 1\}^*, M \text{ also loops on } ww\}$.

Let's continue the analysis from last week:

- Prove that there is no mapping reduction from X_1 to X_3 .
- Prove that $\text{ALL}_{\text{TM}} \leq_m X_4$.
- Prove that $X_4 \leq_m \overline{X_2}$.
- **Extra:** Conclude that there is no mapping reduction $X_4 \leq_m X_2$.